

**THANK
YOU!**



REFERENCES

React Documentation

Frontend JavaScript library for building responsive user interfaces.

<https://react.dev>

Entity Framework Core Documentation

Object-Relational Mapping (ORM) framework for .NET database connectivity.

<https://learn.microsoft.com/ef/core>

JWT (JSON Web Tokens)

Industry-standard authentication tokens for stateless, secure user sessions.

<https://jwt.io>

OpenCV Documentation

Computer vision libraries utilized for biometric face and fingerprint alignment checks.

<https://opencv.org>

Axios Documentation

Promise-based HTTP client for web browser API calls and interceptors.

<https://axios-http.com/docs/intro>

ASP.NET Core Documentation

Backend framework for creating high-performance, cross-platform REST APIs.

<https://learn.microsoft.com/aspnet/core>

MySQL Documentation

Relational database management system for secure and structured transactional storage.

<https://dev.mysql.com/doc/>

Google Authenticator (2FA)

Two-Factor Authentication protocol configuration for clinical identity security.

<https://support.google.com/accounts/answer/...>

Swagger / OpenAPI

Interactive documentation and testing interface for Hospital API endpoints.

<https://swagger.io/docs/>

Microsoft Learn - C# Guide

Object-oriented programming language documentation used for backend HIMS logic.

<https://learn.microsoft.com/dotnet/csharp/>

FEASIBILITY

The **Hospital Information Management System (HIMS)** is technically and operationally feasible as it is developed using reliable and widely adopted technologies such as **React.js, ASP.NET Core Web API, C#, Entity Framework Core, MySQL, JWT Authentication, Biometric Face & Fingerprint Verification, and RESTful APIs**. The system can be deployed using standard hardware and software infrastructure, requires minimal maintenance, and provides a secure, HIPAA-compliant, and modular architecture suitable for modern healthcare organizations.

VIABILITY

The **Hospital Information Management System (HIMS)** is a viable solution because it streamlines clinical and administrative lifecycles by reducing patient registration time, preventing duplicate medical records through intelligent deduplication, ensuring secure access via dual-factor biometrics, and facilitating automated pharmacy and billing processes. These features enhance patient care quality and provider efficiency while reducing administrative operational costs, making the system highly practical, sustainable, and beneficial for long-term clinical use.

HOSPITAL INFORMATION MANAGEMENT SYSTEM

TECHNICAL STACK & SYSTEM ARCHITECTURE

TECHNICAL STACK

FRONTEND

- React.js (v18)
- HTML5 & CSS3
- TailwindCSS

BACKEND

- ASP.NET Core API
- Entity Framework
- C# 12

DATABASE

- MySQL Relational DB

AUTH & SECURITY

- JWT Auth
- Google 2FA

AI & ML FEATURES

- Levenshtein Fuzzy
- Face Embed Match

TOOLS

- VS Code & Git
- Swagger / OpenAPI

SYSTEM ARCHITECTURE

CLIENT LAYER

Patient Portal

Staff Console

Doctor Dashboard

Responsive UI

PRESENTATION

React.js Web App

HTML5 / CSS3 / Axios

APPLICATION LAYER

ASP.NET Web API

- User & Patient Records
- Doctor Scheduling
- Billing & Pharmacy
- Lab Reports & Audits

DATA LAYER

MySQL Database

- Patient & Doctor DB
- Billing & Prescr. DB
- System Settings DB
- Compliance Audit DB

SERVICES

Email (SMTP)

SMS Service

Google 2FA

Biometric API

Cross Cutting Concerns (Project Specific)

Auth & JWT

Input Valid

Error Handle

HIPAA Audit

Data Encrypt

File Sec

Session Mgmt

Alerts

INFRASTRUCTURE & DEPLOYMENT FLOW



Code Dev



GitHub



Build



Test



Deploy



Kestrel



MySQL

ARCHITECTURE HIGHLIGHTS

- ✓ Layered N-Tier Architecture
- ✓ Secure JWT + Biometrics
- ✓ HIPAA Security Audit Logs
- ✓ AI Patient Deduplication
- ✓ Centralised Lab & Pharmacy
- ✓ Role-Based Clinical Control

HOSPITAL INFORMATION MANAGEMENT SYSTEM

IMPACT & BENEFITS

System Impact



Faster Patient Onboarding

Automates registration and scheduling, reducing client wait times.



HIPAA Security & Compliance

Ensures medical records are protected through JWT, 2FA, and biometrics.



Deduplicated Health Records

Uses Levenshtein algorithms to resolve record duplications automatically.



Paperless Clinical Records

Secures digitised record storage, billing audits, and inventory management.



Reduced Administrative Costs

Eliminates manual entry errors and speeds up clinic decision loops.

Benefits Realization



FOR PATIENTS

- ▶ Easy online registration & profile lookup
- ▶ Real-time appointment alerts
- ▶ Secure login to clinical histories
- ▶ Accurate billing & payment receipts



FOR HOSPITAL & CLINICAL PROVIDERS

- ▶ Unified medical history view per patient
- ▶ Biometric patient check-in checks
- ▶ Centralised lab & pharmacy inventory
- ▶ HIPAA compliant logs and audit trails



TECHNICAL BENEFITS

- ▶ Secure RESTful API design
- ▶ Role-based access controls (RBAC)
- ▶ Centralised MySQL database
- ▶ Responsive React.js & Vite build